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STAFF REPORT FOR THE 2019 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS[1]

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Risk of external debt distress:	Moderate
Overall risk of debt distress:	Moderate
Granularity in the risk rating:	Limited space to absorb shock
Application of judgement:	No

The updated DSA suggests that the external risk of debt distress for Vanuatu remains moderate with limited space to absorb shocks. All external debt indicators remain below the relevant indicative thresholds under the baseline scenario, incorporating the average long-term effects of natural disasters on growth and the fiscal and current account balances. A tailored natural disaster shock, reflecting Vanuatu's vulnerability to disasters, would cause the present value (PV) of public and publicly guaranteed (PPG) external debt-to-GDP ratio to breach the threshold from 2024 onwards.

The overall risk of debt distress is assessed as moderate. Although the PV of the public-debt-to-GDP ratio remains below the 55 percent benchmark under the baseline scenario, the public-debt-to-GDP ratio would breach the authorities' debt ceiling of 60 percent by 2025. Moreover, a tailored natural disaster shock would lead to a significant deterioration in debt sustainability, breaching the benchmark. The breach of the authorities' debt ceiling and of the benchmark indicates the need for rebuilding fiscal buffers and enhancing resilience against shocks, including from natural disasters. This requires both stronger revenue mobilization measures, including an introduction of the proposed income taxes, and expenditure rationalization in the medium term. When contracting new public infrastructure projects, the authorities are encouraged to seek grants or concessional loans as much as possible to contain its debt burden.

[1] The two most recent observations for Vanuatu's Composite Indicator (CI) index are 2.94 and 2.99, indicating that the country's debt-carrying capacity is medium. According to the new "Guidance Note on the Bank-Fund Debt Sustainability Framework for Low-income Countries" (http://www.imf.org/en/Publications/Policy-Papers/Issues/2018/02/14/pp122617guidance-note-on-lic-dsf), the relevant indicative thresholds for the medium category are: 40 percent for the present value (PV) of the debt-to-GDP ratio,180 percent for the PV of the debt-to-exports ratio, 15 percent for the debt service-to-exports ratio, and 18 percent for the debt-service-to-revenue ratio. These thresholds are applicable to public and publicly guaranteed (PPG) external debt. The benchmark of the PV of total public-sector debt for a medium debt carrying capacity is 55 percent.

PUBLIC DEBT COVERAGE

1. The coverage of public sector debt for this debt sustainability analysis is central government debt, central government-guaranteed debt, and central bank debt, which has been borrowed on behalf of the government. [2] Because of data limitations, non-guaranteed SOE debt and private external debt are not included in the analysis.[3] Given the limited capacity to borrow both externally and domestically by Vanuatu's state and local governments, SOEs and its private sector, data deficiencies do not affect the overall assessment.

Coverage of Public Sector Debt

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	X
8	Non-guaranteed SOE debt	

BACKGROUND ON DEBT

2. After Cyclone Pam struck Vanuatu in 2015, public sector debt has increased sharply to 52.4 percent of GDP in 2018 from 26.1 percent in 2014. This was mainly caused by new disbursement for infrastructure development supported by bilateral partners, including the Japan International Cooperation Agency (JICA) and the Export-Import Bank of China (China EXIM Bank). In addition to the IMF's disbursement of USD 23.8 million in June 2015, the IDA and ADB have provided loans and grants to support the reconstruction and improvement of

Stock of Public Debt (External and Domestic) at End-2018

	In million of Vatu	In million of US dollars	As a share of total debt	In percent of GDP
Total public debt	52,826	471	100.0	52.4
External	45,520	406	86.2	45.2
Multilateral	16,466	147	31.2	16.3
ADB	7,726	69	14.6	7.7
IDA	6,478	58	12.3	6.4
IMF	2,262	20	4.3	2.2
Bilateral	27,332	243	51.7	27.1
China EXIM Bank	17,080	152	32.3	17.0
JICA	10,201	91	19.3	10.1
Others	51	0	0.1	0.1
Publicly guaranteed debt	1,722	15	3.3	1.7
Domestic	7,307	65	13.8	7.3
Government bonds	6,267	56	11.9	6.2
RBV	2,513	22	4.8	2.5
Public Corporation	2,350	21	4.4	2.3
Commercial Banks	1,288	11	2.4	1.3
Others	116	1	0.2	0.1
Publicly guaranteed debt	1,040	9	2.0	1.0

Source: Vanuatu authorities and IMF staff estimates.

[2] The technical assistance provided by PFTAC helps the country's authorities expand the coverage of government financial statistics (GFS) from budgetary central government to general government. The broader coverage of public sector debt can be expected as a result of the PFTAC TA program. For non-guaranteed SOE debt, the authorities are currently revising the *Government Business Enterprises Act* bill, which will be complemented by the upcoming report that will discuss the financial position of, and fiscal risks posed by SOEs. The report should enable staff to identifiy or estimate the size of the SOE debt.

[3] Please note that the size of private external debt does not affect the risk rating for PPG external debt and public-sector debt, which does not include private external debt.

roads and schools. As of end-2018, the share of bilateral and multilateral creditors amounted to 51.7 and 31.2 percent of total public debt, respectively. Of public domestic debt, central government bonds were largely held by public corporations (primarily the Vanuatu National Provident Fund, VNPF), followed by the Reserve Bank of Vanuatu (RBV) and commercial banks. There are also government-guaranteed debts for state-owned enterprises (SOEs), such as Air Vanuatu prior to 2019.[4]

3. Windfall revenues from the economic citizenship programs (ECPs) have enabled the authorities to embark on a debt reduction program.[5] They paid off domestic and external debt in the amount of VUV 1.8 billion and VUV 1.5 billion, respectively, in 2018. External loan repayments included VUV 1.0 billion to China, VUV 0.4 billion to the ADB, and VUV 60 million to the IDA.

4. Following the end of the Tanna and Malekula Road Rehabilitation and Upgrade Program (Phase I), the authorities signed a Phase II loan agreement, amounting to VUV 5.7 billion, with China in November 2018. As the Phase II project was effectively an extension of the Phase I project, the financing terms and grant element were the same.[6] The grant element was 29.2 percent, lower than the authorities' commitment of a 35 percent grant component, which was first introduced in 2015 under its *Debt Management Strategy (2015–17)*, one year after the Phase I project was signed. The authorities intend to retain their 35 percent grant component target as they update the debt management strategy.

BACKGROUND ON MACROECONOMIC FORECASTS

5. Similar to the last DSA, the baseline scenario, which is consistent with the macroeconomic framework, incorporates the effects of natural disasters and climate change over the longer-term. The years 2019–24 are assumed to be free from newly-occurring major, costly disasters to simplify the policy discussion of the near-term outlook - standard practice in DSAs for other Pacific island small states with a similar risk profile. However, from 2025 onwards, the baseline incorporates the average long-term effects of natural disasters and climate change. Based on staff's research on the impact of natural disasters, real GDP growth is lowered by 0.5 percentage points annually, the current account deficit is raised by 1.3 percentage points of GDP and the fiscal deficit is increased by 0.35 percentage points of GDP relative to disaster-free projections.[7] The projected changes in 2025 for these three variables are smaller than the effects listed above. This is because real GDP growth is projected to rise in 2025 (in the absence of newly-occurring natural disasters), as the negative effect of recent natural disasters such as the Ambae and Ambryn volcanic

[4] The amounts are the original guarantees as provided by the authorities. The current outstanding amount might be lower, if some of the guaranteed debt has been repaid by the debt's issuer.

[5] The ECPs include the Vanuatu Development Support Program (VDSP) and Vanuatu Contribution Program (VCP), outlined in Box 1 of Vanuatu: 2018 Article IV Consultation, IMF Country Report No. 18/109.

[6] The financing terms of China's new loan are an interest rate of 2 percent, a grace period which shall not exceed 7 years, and a maturity of 20 years. A grant element has been calculated based on financing terms with a 5 percent discount rate.

[7] Please see the detail in the, Lee and others, 2018, "The Economic Impact of Natural Disaster in Pacific Island Countries," IMF Working Paper 18/108 (https://www.imf.org/en/Publications/WP/Issues/2018/05/10/The-Economic-Impact-of-Natural-Disasters-in-Pacific-Island-Countries-Adaptation-and-45826).

eruptions and Cyclone Oma wanes. Similarly, the assumed increase in the fiscal and current account deficits from 2025 onwards (due to incorporation of the average effect of natural disasters and climate change) is masked somewhat by a coincident decline in the projected acquisition of non-financial assets. The discount rate used to calculate the net present value of external debt remains at 5 percent. The main assumptions are:

- ***Real GDP growth*** is projected at 2.8 percent on average during 2019–29, which is lower than growth rates in the past three years, during which public investment has boomed and reconstruction efforts have been ongoing after Cyclone Pam. It is also lower than the 10-year average of 3.1 percent in the previous DSA, better reflecting the authorities' longstanding views.
- ***Inflation*** (measured by the GDP price deflator) is projected to average 2.2 percent (in U.S. dollar terms, the relevant measure for external debt), and 2.5 percent (in domestic currency term, the relevant measure for public debt) during 2019–29 both similar to their historical averages.
- The ***non-interest current account deficit*** is projected to rise to 3.8 percent of GDP on average over 2019–29, relative to the historical average of 2.7 percent. This reflects the high import content for key infrastructure projects. It is lower than last year's assumption of 6.6 percent because of higher expected ECP revenues and remittances.
- ***Foreign direct investment inflows*** are expected to average 3.3 percent of GDP over 2019–29, lower than the historical average of 5.1 percent, which included the post-Cyclone-Pam investment boom.
- The ***primary deficit*** is expected to be 3.1 percent of GDP on average over 2019–2029, more negative than the historical average of 1.1 percent and last year's assumption of 2.5 percent. This reflects higher infrastructure spending going forward than the previous DSA, and a more accurate accounting of maintenance costs for infrastructure. Staff does not take into account the possible introduction of income taxes.
- ***External borrowing and grants*** will continue to be strong. Borrowing is driven in the short term by the disbursements for the new USD 51 million project supported by China, that should take place from 2019 to 2021. From 2022 onwards, staff assumes a slightly lower disbursement as a percent of GDP from bilateral development partners. The level of new annual external borrowing is expected to average around 5.0 percent of GDP, higher than last year's assumption of 4.2 percent of GDP, reflecting the new loan from China and the authorities' increased willingness to borrow externally to promote infrastructure development. Grant and lending flows from multilateral development partners are expected to increase over the medium term because of the scaling-up of IDA and ADB financing.[8] However, grants are expected to decline over the longer term, as the country's economy grows.

[8] With respect to projected new borrowing from IDA and ADB, DSAs always assume terms that would prevail if the country was at low risk of debt distress, independent of current actual terms (which can change on a year to year basis). This is done to avoid a circular situation where the assumption that future commitments will be on grant terms would yield actual commitments on credit terms.

- The **government-guaranteed debts** as of end-2018 will continue for the projection period. Staff assumed that the government would not provide any guarantees for any new borrowing by SOEs, including Air Vanuatu.

6. The realism tool highlights that assumptions on the primary balance are conservative (Figure 4). The three-year adjustment in the primary deficit between 2018 and 2021 is at 8.9 percent of GDP. The deteriorating fiscal position is based on a conservative assumption for the proceeds from the ECPs and stronger infrastructure spending. The assumption on real growth in 2019 and 2020 is lower than possible growth paths which are calculated by the model based on a one-year fiscal adjustment. Two charts on public and private investment rates and their contributions to real GDP growth are not available because of a lack of data. Staff will try to estimate the capital stock if relevant information becomes available.

Vanuatu: Baseline Macroeconomic Assumption

(In percent of GDP, unless otherwise stated)

	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2008-18 Historical average	2019-29 average	2018 DSA 2018-28 average	Effect of natural disaster and climate change
Real GDP growth	3.4	3.0	2.8	2.8	2.9	2.9	2.6	2.6	2.6	2.6	2.6	2.3	2.8	3.1	0.5
GDP deflator in US dollars (change in percent)	-1.3	2.3	2.5	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.2	2.2	3.5	
Non-interest current account deficit	0.7	4.9	4.2	3.8	3.5	3.6	4.1	4.2	4.2	4.3	4.4	2.7	3.8	6.6	1.3
Net FDI (negative = inflow)	-4.2	-4.0	-3.8	-3.6	-3.4	-3.2	-3.1	-2.9	-2.8	-2.6	-2.5	-5.1	-3.3	-3.6	
Primary deficit	2.3	2.7	3.1	3.2	3.2	3.1	3.2	3.2	3.2	3.2	3.2	1.1	3.1	2.5	0.35
Grants	5.8	5.8	5.7	5.6	5.6	5.5	5.5	5.5	5.5	5.6	5.6	6.1	5.6	7.6	

Source: IMF staff projections.

COUNTRY CLASSIFICATION

7. The country's debt-carrying capacity as applied in the 2019 DSA is medium. The Composite Indicator (CI) index for Vanuatu, which has been calculated based on the April 2019 WEO (but with updated data on remittances) and the 2017 CPIA, is 2.94, indicating that the county's debt-carrying capacity would be medium in the revised LIC-DSF framework.

Calculation of the CI Index

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.385	3.388	1.30	44%
Real growth rate (in percent)	2.719	2.840	0.08	3%
Import coverage of reserves (in percent)	4.052	54.466	2.21	74%
Import coverage of reserves^2 (in percent)	-3.990	29.665	-1.18	-40%
Remittances as percent of GDP (in percent)	2.022	4.846	0.10	3%
World economic growth (in percent)	13.520	3.579	0.48	16%
CI Score			2.99	100%
CI rating			Medium	

Source: IMF staff calculations.

8. The relevant indicative thresholds for the medium category are: 40 percent for the PV of debt-to-GDP ratio,180 percent for the PV of the debt-to-exports ratio, 15 percent for the debt service-to-exports ratio, and 18 percent for the debt service-to-revenue ratio. These thresholds are applicable to public and publicly guaranteed (PPG) external debt. The benchmark for the PV of total public sector debt under medium debt carrying capacity is 55 percent of GDP.

PPG External Debt Thresholds and Total Public Debt Benchmarks

Debt carrying capacity (CI classification)	PV of PPG external debt in percent of		PV of PPG external debt in percent of		PV of total public debt in percent of
	GDP	Exports	Exports	Revenue	GDP
Weak	30	140	10	14	35
Medium	40	180	15	18	55
Strong	55	240	21	23	70

DETERMINATION OF SCENARIO STRESS TESTS

9. Given Vanuatu's vulnerability to natural disasters, staff conducted a tailored stress test for a natural disaster shock. Vanuatu, which is defined as a small developing natural-disaster-prone state in the IMF (2016) policy paper on small states, is automatically subject to the LIC-DSF standard natural disaster shock. This is a one-off shock of 10 percentage points to the debt-to-GDP ratio in the second year of the projection period (2020 in this case). Staff adjusted the default parameters by assuming a reduction of real GDP and export growth by 4 and 10 percentage points respectively.[9] For combined contingent liability shock, staff adjusted the levels for the increase in public debt from SOEs from 2 percent to 4 percent of GDP to reflect the government's financial support to Air Vanuatu in 2019.[10] Staff continued using the default decrease in GDP of 5 percent from financial market turbulence. For Vanuatu, the default 5 percent of GDP value of the contingent liability can be interpreted as including a capital injection to an undercapitalized domestic bank.

Combined Contingent Liability Shock

1 **The country's coverage of public debt**	The central government, central bank, government-guaranteed debt		
	Default	**Used for the analysis**	**Reasons for deviations from the default settings**
2 Other elements of the general government not captured in 1.	0 percent of GDP	0.0	
3 SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	4.0	To reflect the government's financial support to Air Vanuatu
4 PPP	35 percent of PPP stock	0.0	
5 Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5.0	
Total (2+3+4+5) (in percent of GDP)		9.0	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

[9] Please see the details in IMF, 2016, "Small States' Resilience to Natural Disasters and Climate Change: Role for the IMF," *IMF Policy Paper December 2016* (https://www.imf.org/en/Publications/Policy-Papers/Issues/2016/12/31/Small-States-Resilience-to-Natural-Disasters-and-Climate-Change-Role-for-the-IMF-PP5079).

[10] In February and April 2019, the government provided two loans to Air Vanuatu, totaling VUV 1,230 million. The money is being used as deposits to purchase new aircraft. Further costs associated with these aircraft purchases have not been incorporated into this DSA because of a lack of confirmed information on the terms and potential financing sources.

DEBT SUSTAINABILITY

A. External Debt Sustainability Analysis

10. All external PPG debt indicators remain below the policy-relevant thresholds for the projection period under the baseline scenario (Figure 1 and Table 1). These thresholds include the present value (PV) of the external-debt-to-GDP ratio, the PV of the external-debt-to-exports ratio, the external-debt-service-to-exports ratio, and the external-debt-service-to-revenue ratio. The PV of external-debt-to GDP ratio is expected to increase continuously from 27.0 percent in 2018 to 35.6 percent in 2029 mainly because of new disbursements for key infrastructure projects. As Figure 3 shows, the main driver of debt dynamics during the projection period is the current account deficit.

11. The stress tests indicate that a tailored natural disaster shock has the largest impact on debt trajectory, causing a breach of the external-debt-to-GDP threshold from 2024 onwards. This suggests the need for rebuilding fiscal buffers to enhance resilience against natural disasters. Other tests, including shocks to exports, other flows and the nominal exchange rate (depreciation), would also lead to breaches in the thresholds (Table 3). The export shock, which was the largest impact under the 2018 DSA, still is the fourth largest impact, which continues to suggest the need for expanding the export base through economic diversification.

B. Public Sector Debt Sustainability Analysis

12. The PV of public-debt-to-GDP ratio does not breach the 55 percent benchmark under the baseline scenario (Figure 2 and Table 2). However, the public-debt-to-GDP ratio would rise from 52.4 percent in 2018 to breach the authorities' stated public-debt-to-GDP target of 60 percent by 2025 (Table 2). As Figure 3 indicates, the breach is primarily driven by a primary deficit caused by elevated capital spending.

13. The stress tests, including the contingent liability shock, demonstrate deteriorating debt sustainability (Figure 2 and Table 4). The tailored natural disaster shock would breach its benchmark in 2029, while other shocks would not result in a breach (Table 4). The shock to real GDP growth has the third largest impact on debt sustainability. This highlights the need for encouraging stronger economic growth and the importance of rebuilding fiscal buffers against external shocks in the medium term. The contingent liability shock would not lead to a breach, but would result in an average deterioration of the debt position relative to the baseline of 5 percent of GDP. The authorities need to consider fiscal risk from contingent liability across SOEs when they provide guarantees to them.

RISK RATING AND VULNERABILITIES

14. The debt sustainability analysis under the revised LIC-DSF framework suggests that Vanuatu's risk of external debt distress remains moderate, with limited space to absorb shocks. While there is no breach of external debt thresholds under the baseline scenario, the results of the stress tests indicate that the tailored natural disaster shock would result in a breach of the

threshold for the PV of external-debt-to-GDP ratio. This underscores the importance of enhancing resilience against natural disasters. Figure 5 shows that there is limited space to absorb shocks, indicating the need for creating fiscal space to address future shocks. Even though debt service indicators remain well below their thresholds both under the baseline and stress test scenarios, loan-funded projects should be contracted as much as possible on favorable concessional terms to help contain the debt burden and respect the authorities' stated goal of achieving a 35 percent grant element for such loans.

15. The DSA suggests that overall risk of debt distress is moderate. Even though the 55 percent benchmark for the PV of the external-debt-to-GDP ratio would not be breached under the baseline scenario, the public-debt-to-GDP ratio would breach the authorities' target of 60 percent by 2025. This suggests the need for both stronger revenue mobilization measures, including the introduction of an income tax (personal and/or corporate), and expenditure rationalization in the medium term. Additional borrowing from other bilateral partners and a provision of additional debt guarantees would result in a breach of the authorities' target in the short term. The authorities need to prioritize which loans to accept and limit guarantees to SOEs, including Air Vanuatu, to safeguard debt sustainability. The tailored natural disaster shock has the largest impact on public debt sustainability, resulting in the PV of external-debt-to-GDP ratio reaching 55 percent in 2029. The authorities are encouraged to rebuild fiscal buffers to enhance resilience against external shocks.

AUTHORITIES' VIEWS

16. The authorities broadly agreed with the staff assessment of debt sustainability analysis under the revised LIC-DSF. Given high infrastructure needs, the authorities underscored the need for financial support from bilateral and multilateral donors for any new projects. At the same time, they intended to maintain a grant-element target of at least 35 percent and seek grant financing as much as possible to reduce debt burden. They stressed their commitment to make prepayments to contain debt accumulation by using their strong cash reserves, which had been accumulating from particularly strong ECP revenues starting in 2016. They noted a difference in the coverage of public sector debt relative to published *Budget 2019* figures, which did not include the IMF loans disbursed after Cyclone Pam in 2015 in their definition of public debt, as the payments had been directed to the RBV. The difference is very limited at 2.2 percent of GDP as of end-2018. The authorities noted that the DSA uses the original guaranteed debt amounts provided by the authorities.

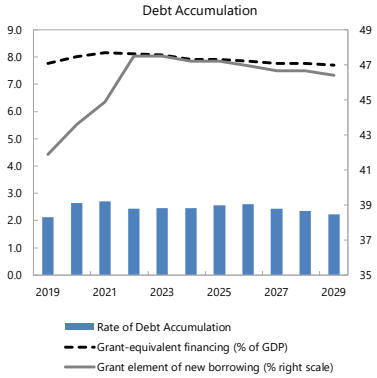
17. The authorities also agreed with the assumptions in the DSA used in its forecasts. This included assumptions on the grant element of new loans and emphasis on the PPG debt target, as they remained committed to maintaining a grant-element target of at least 35 percent on new loans and the PPG-debt-to-GDP target of 60 percent. The DSA's approach to the forecasts for publicly-guaranteed debt matched the government's strongly stressed intention that it will be difficult to provide any guarantees in the near future for borrowing by SOEs.

Table 1. Vanuatu: External Debt Sustainability Framework, Baseline Scenario, 2016–39

(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2016	2017	2018	2019	2020	2021	2022	2023	2024	2029	2039	Historical	Projections
External debt (nominal) 1/	**36.2**	**43.8**	**45.2**	**46.0**	**47.5**	**49.2**	**50.6**	**51.9**	**53.0**	**57.2**	**54.0**	**19.8**	**52.5**
of which: public and publicly guaranteed (PPG)	**36.2**	**43.8**	**45.2**	**46.0**	**47.5**	**49.2**	**50.6**	**51.9**	**53.0**	**57.2**	**54.0**	**19.8**	**52.5**
Change in external debt	9.1	7.5	1.4	0.8	1.5	1.7	1.4	1.2	1.1	0.4	-0.6		
Identified net debt-creating flows	**-6.9**	**-1.4**	**-10.1**	**-4.5**	**0.1**	**-0.3**	**-0.5**	**-0.7**	**-0.5**	**1.2**	**0.4**	**-4.9**	**-0.2**
Non-interest current account deficit	**-0.8**	**6.1**	**-4.0**	**0.7**	**4.9**	**4.2**	**3.8**	**3.5**	**3.6**	**4.4**	**2.6**	**2.7**	**3.8**
Deficit in balance of goods and services	15.4	12.2	9.7	7.1	10.9	10.0	9.2	8.7	8.5	8.1	4.6	**9.0**	**8.8**
Exports	43.2	41.5	44.3	46.8	46.5	46.2	45.8	45.5	45.3	44.8	44.9		
Imports	58.6	53.7	54.1	53.9	57.4	56.2	55.0	54.2	53.7	52.9	49.5		
Net current transfers (negative = inflow)	-16.1	-7.0	-12.3	-4.9	-4.6	-4.3	-4.0	-3.9	-3.9	-3.5	-3.0	**-7.6**	**-4.0**
of which: official	-10.9	-1.8	-2.2	-1.8	-1.7	-1.6	-1.5	-1.5	-1.4	-1.1	-0.6		
Other current account flows (negative = net inflow)	0.0	0.8	-1.4	-1.4	-1.4	-1.4	-1.3	-1.2	-1.0	-0.2	1.1	**1.3**	**-1.0**
Net FDI (negative = inflow)	**-5.6**	**-4.5**	**-4.4**	**-4.2**	**-4.0**	**-3.8**	**-3.6**	**-3.4**	**-3.2**	**-2.5**	**-1.5**	**-5.1**	**-3.3**
Endogenous debt dynamics 2/	**-0.5**	**-3.0**	**-1.8**	**-1.0**	**-0.8**	**-0.7**	**-0.7**	**-0.8**	**-0.8**	**-0.7**	**-0.7**		
Contribution from nominal interest rate	0.3	0.4	0.5	0.5	0.5	0.5	0.6	0.6	0.6	0.7	0.7		
Contribution from real GDP growth	-0.9	-1.4	-1.3	-1.5	-1.3	-1.3	-1.3	-1.4	-1.4	-1.4	-1.4		
Contribution from price and exchange rate changes	0.1	-1.9	-0.9	…	…	…	…	…	…	…	…		
Residual 3/	**16.0**	**8.9**	**11.5**	**5.3**	**1.5**	**2.0**	**1.9**	**1.9**	**1.6**	**-0.7**	**-1.1**	**10.2**	**1.3**
of which: exceptional financing	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**…**	**…**	**27.0**	**28.5**	**29.5**	**30.6**	**31.3**	**32.0**	**32.7**	**35.6**	**34.4**		
PV of PPG external debt-to-exports ratio	**…**	**…**	**60.8**	**60.9**	**63.5**	**66.3**	**68.4**	**70.3**	**72.1**	**79.6**	**76.7**		
PPG debt service-to-exports ratio	**2.1**	**2.3**	**5.1**	**4.1**	**3.6**	**3.9**	**4.1**	**4.2**	**4.1**	**4.5**	**5.1**		
PPG debt service-to-revenue ratio	**4.1**	**3.6**	**7.6**	**8.6**	**7.5**	**8.2**	**8.5**	**8.6**	**8.4**	**9.6**	**11.1**		
Gross external financing need (Million of U.S. dollars)	-43.2	22.5	-56.9	-14.9	25.2	23.7	22.9	23.5	27.0	62.7	91.4		
Key macroeconomic assumptions													
Real GDP growth (in percent)	3.5	4.4	3.2	3.4	3.0	2.8	2.8	2.9	2.9	2.6	2.6	**2.3**	**2.8**
GDP deflator in US dollar terms (change in percent)	-0.3	5.6	2.1	-1.3	2.3	2.5	2.6	2.6	2.6	2.6	2.6	**2.2**	**2.2**
Effective interest rate (percent) 4/	1.1	1.2	1.2	1.1	1.2	1.2	1.3	1.3	1.3	1.3	1.3	**1.4**	**1.2**
Growth of exports of G&S (US dollar terms, in percent)	5.0	5.7	12.8	7.7	4.7	4.6	4.6	4.9	4.9	5.1	5.3	**4.6**	**5.2**
Growth of imports of G&S (US dollar terms, in percent)	-6.0	1.0	6.2	1.7	12.3	3.1	3.2	4.1	4.7	4.9	2.9	**3.3**	**4.9**
Grant element of new public sector borrowing (in percent)	…	…	…	41.9	43.6	44.9	47.5	47.5	47.2	46.4	44.5	**…**	**46.0**
Government revenues (excluding grants, in percent of GDP)	22.5	26.4	29.9	22.1	22.1	22.2	22.2	22.1	22.0	21.3	20.5	**21.1**	**21.9**
Aid flows (in Million of US dollars) 5/	66.2	74.1	52.4	82.5	91.2	99.1	107.2	112.4	115.9	143.5	218.6		
Grant-equivalent financing (in percent of GDP) 6/	…	…	…	7.8	8.0	8.2	8.1	8.1	7.9	7.7	7.3	**…**	**7.9**
Grant-equivalent financing (in percent of external financing) 6/	…	…	…	74.5	73.9	73.2	74.7	74.8	74.8	75.7	78.0	**…**	**74.8**
Nominal GDP (Million of US dollars)	798	880	928	947	998	1051	1109	1170	1236	1598	2675		
Nominal dollar GDP growth	3.1	10.3	5.4	2.1	5.4	5.3	5.5	5.6	5.6	5.3	5.3	**4.6**	**5.1**
Memorandum items:													
PV of external debt 7/	…	…	27.0	28.5	29.5	30.6	31.3	32.0	32.7	35.6	34.4		
In percent of exports	…	…	60.8	60.9	63.5	66.3	68.4	70.3	72.1	79.6	76.7		
Total external debt service-to-exports ratio	2.1	2.3	5.1	4.1	3.6	3.9	4.1	4.2	4.1	4.5	5.1		
PV of PPG external debt (in Million of US dollars)			250.2	269.9	295.0	321.9	347.6	374.7	403.5	569.7	920.3		
(PVt-PVt-1)/GDPt-1 (in percent)				2.1	2.6	2.7	2.4	2.4	2.5	2.2	1.5		
Non-interest current account deficit that stabilizes debt ratio	-9.9	-1.5	-5.4	0.0	3.3	2.5	2.4	2.3	2.5	4.0	3.3		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	No



Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + Ɛα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, Ɛ=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

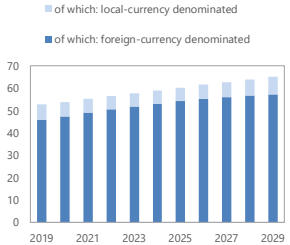
Table 2. Vanuatu: Public Sector Debt Sustainability Framework, Baseline Scenario, 2016–39

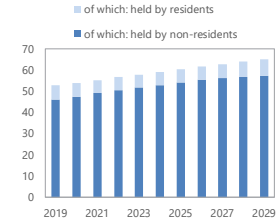
(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2016	2017	2018	2019	2020	2021	2022	2023	2024	2029	2039	Historical	Projections
Public sector debt 1/	**46.4**	**53.2**	**52.4**	**52.9**	**53.9**	**55.3**	**56.6**	**57.9**	**59.0**	**65.1**	**74.4**	**27.6**	**59.0**
of which: external debt	**36.2**	**43.8**	**45.2**	**46.0**	**47.5**	**49.2**	**50.6**	**51.9**	**53.0**	**57.2**	**54.0**	**19.8**	**52.5**
Change in public sector debt	10.5	6.8	-0.7	0.5	1.0	1.4	1.3	1.2	1.1	1.1	0.8		
Identified debt-creating flows	**2.9**	**-4.8**	**-5.6**	**0.5**	**1.0**	**1.4**	**1.3**	**1.2**	**1.1**	**1.1**	**0.8**	**0.5**	**1.2**
Primary deficit	**2.9**	**-0.1**	**-5.8**	**2.3**	**2.7**	**3.1**	**3.2**	**3.2**	**3.1**	**3.2**	**2.6**	**1.1**	**3.1**
Revenue and grants	30.8	34.8	35.5	27.9	28.0	28.0	27.8	27.7	27.6	26.8	26.2	**27.3**	**27.5**
of which: grants	8.3	8.4	5.6	5.8	5.8	5.7	5.6	5.6	5.5	5.6	5.6		
Primary (noninterest) expenditure	33.7	34.7	29.7	30.2	30.7	31.1	31.0	30.9	30.7	30.0	28.7	**28.3**	**30.5**
Automatic debt dynamics	**0.0**	**-4.6**	**0.2**	**-1.8**	**-1.7**	**-1.7**	**-1.8**	**-1.9**	**-2.0**	**-2.0**	**-1.8**		
Contribution from interest rate/growth differential	-0.8	-1.7	-1.4	-1.7	-1.5	-1.6	-1.5	-1.6	-1.7	-1.7	-1.5		
of which: contribution from average real interest rate	0.4	0.3	0.2	0.0	0.0	-0.1	0.0	0.0	-0.1	-0.1	0.3		
of which: contribution from real GDP growth	-1.2	-2.0	-1.6	-1.7	-1.5	-1.5	-1.5	-1.6	-1.6	-1.6	-1.9		
Contribution from real exchange rate depreciation	0.8	-3.0	1.6	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**7.6**	**11.6**	**4.9**	**-0.1**	**-0.2**	**-0.2**	**-0.3**	**-0.3**	**-0.3**	**-0.3**	**-0.3**	**4.7**	**-0.2**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**35.1**	**35.4**	**36.0**	**36.7**	**37.4**	**38.0**	**38.7**	**43.6**	**54.8**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**99.0**	**126.8**	**128.6**	**131.2**	**134.4**	**137.2**	**140.4**	**162.4**	**209.4**		
Debt service-to-revenue and grants ratio 3/	**9.9**	**9.8**	**12.4**	**10.8**	**12.2**	**10.9**	**11.7**	**11.7**	**8.6**	**11.7**	**16.7**		
Gross financing need 4/	6.0	3.3	-1.4	5.3	6.1	6.2	6.4	6.4	5.5	6.3	7.0		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	3.5	4.4	3.2	3.4	3.0	2.8	2.8	2.9	2.9	2.6	2.6	**2.3**	**2.8**
Average nominal interest rate on external debt (in percent)	1.2	1.2	1.2	1.1	1.2	1.2	1.3	1.3	1.3	1.3	1.3	**1.4**	**1.2**
Average real interest rate on domestic debt (in percent)	7.0	2.5	3.2	4.7	4.7	4.8	4.7	4.8	4.8	4.4	4.1	**3.8**	**4.7**
Real exchange rate depreciation (in percent, + indicates depreciation)	3.1	-8.6	3.8	...	...	...	...	...	...	...	...	**-0.9**	**...**
Inflation rate (GDP deflator, in percent)	1.8	4.2	2.9	2.1	2.3	2.5	2.6	2.6	2.6	2.6	2.6	**2.6**	**2.5**
Growth of real primary spending (deflated by GDP deflator, in percent)	-9.2	7.5	-11.7	5.3	4.6	4.3	2.4	2.5	2.2	2.1	2.1	**4.9**	**2.9**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-7.6	-7.0	-5.1	1.8	1.7	1.7	1.8	1.9	2.0	2.0	1.8	**-6.5**	**1.9**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government, central bank, government-guaranteed debt. Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Vanuatu: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2019–29 1/

Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	Yes	
Natural Disasters	No	Yes
Commodity Prices 2/	n.a.	n.a.
Market Financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.3%	1.3%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	31	31
Avg. grace period	8	8

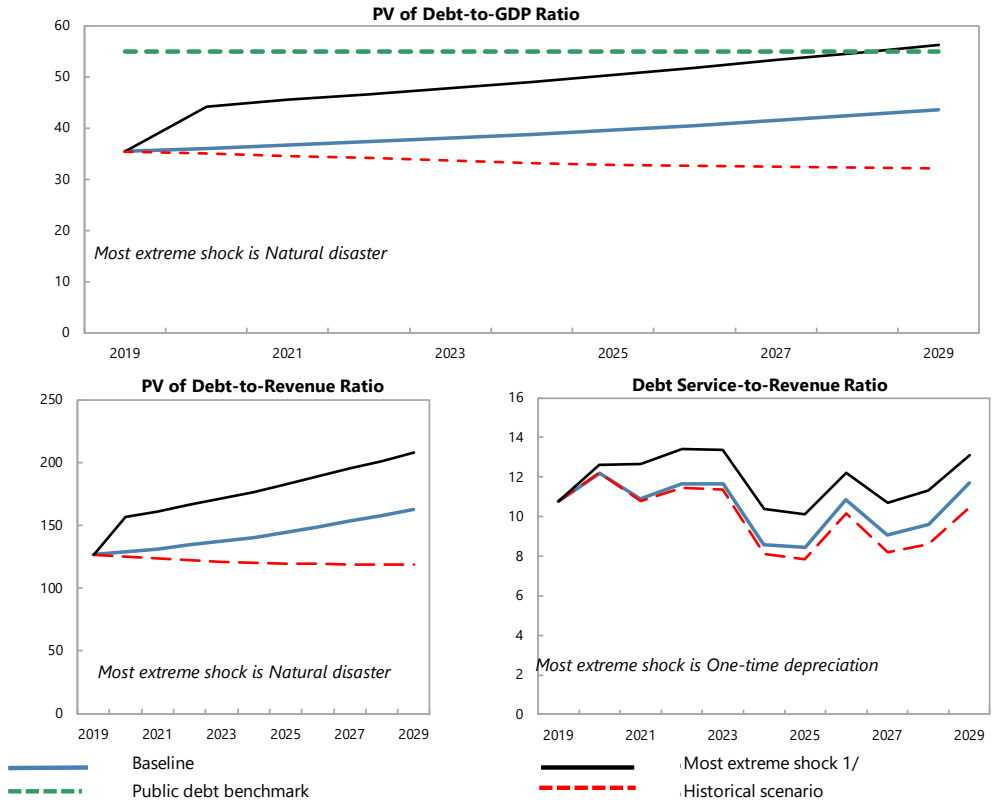
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2029. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Vanuatu: Indicators of Public Debt Under Alternative Scenarios, 2019–29 1/



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	85%	85%
Domestic medium and long-term	15%	15%
Domestic short-term	2%	0%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	1.3%	1.3%
Avg. maturity (incl. grace period)	31	31
Avg. grace period	8	8
Domestic MLT debt		
Avg. real interest rate on new borrowing	4.7%	4.7%
Avg. maturity (incl. grace period)	10	10
Avg. grace period	9	9
Domestic short-term debt		
Avg. real interest rate	-2.5%	-2.5%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2029. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Vanuatu: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2019–29

(In percent)

	Projections 1/										
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
PV of debt-to GDP ratio											
Baseline	28	30	31	31	32	33	33	34	35	35	36
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	28	28	28	27	27	26	25	24	23	21	20
B. Bound Tests											
B1. Real GDP growth	28	30	32	33	34	35	35	36	37	37	38
B2. Primary balance	28	31	34	34	35	36	36	37	38	38	38
B3. Exports	28	32	38	38	39	40	**40**	**41**	**42**	**42**	**42**
B4. Other flows 3/	28	34	**40**	**41**	**41**	**42**	**42**	**43**	**44**	**44**	**44**
B5. Depreciation	28	37	35	36	37	38	39	**40**	**41**	**42**	**42**
B6. Combination of B1-B5	28	35	39	39	40	**41**	**41**	**42**	**43**	**43**	**43**
C. Tailored Tests											
C1. Combined contingent liabilities	28	34	35	36	36	37	38	38	39	39	40
C2. Natural disaster	28	36	37	39	40	**41**	**42**	**43**	**44**	**45**	**46**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	40	40	40	40	40	40	40	40	40	40	40
PV of debt-to-exports ratio											
Baseline	61	64	66	68	70	72	74	76	77	79	80
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	61	60	60	59	59	58	56	54	51	47	44
B. Bound Tests											
B1. Real GDP growth	61	64	66	68	70	72	74	76	77	79	80
B2. Primary balance	61	67	73	75	77	79	81	82	84	85	86
B3. Exports	61	74	94	97	99	101	103	105	106	108	108
B4. Other flows 3/	61	74	87	89	90	92	94	96	97	98	98
B5. Depreciation	61	64	61	63	65	67	69	71	72	74	75
B6. Combination of B1-B5	61	74	77	86	88	90	92	93	95	96	97
C. Tailored Tests											
C1. Combined contingent liabilities	61	73	76	78	80	81	83	85	87	88	89
C2. Natural disaster	61	82	86	90	93	96	99	102	104	107	109
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	180	180	180	180	180	180	180	180	180	180	180
Debt service-to-exports ratio											
Baseline	4	4	4	4	4	4	4	4	4	4	5
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	4	4	4	4	4	4	4	4	4	4	4
B. Bound Tests											
B1. Real GDP growth	4	4	4	4	4	4	4	4	4	4	5
B2. Primary balance	4	4	4	4	4	4	4	4	4	4	5
B3. Exports	4	4	5	5	5	5	5	5	5	5	6
B4. Other flows 3/	4	4	4	5	5	5	4	4	5	5	5
B5. Depreciation	4	4	4	4	4	4	4	4	4	4	4
B6. Combination of B1-B5	4	4	4	5	5	5	5	4	5	5	6
C. Tailored Tests											
C1. Combined contingent liabilities	4	4	4	4	4	4	4	4	4	5	5
C2. Natural disaster	4	4	5	5	5	5	5	5	5	5	5
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	15	15	15	15	15	15	15	15	15	15	15
Debt service-to-revenue ratio											
Baseline	9	8	8	9	9	8	8	8	9	9	10
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	9	8	8	8	8	8	8	7	8	8	8
B. Bound Tests											
B1. Real GDP growth	9	8	9	9	9	9	9	8	9	10	10
B2. Primary balance	9	8	8	9	9	9	8	8	9	9	10
B3. Exports	9	8	9	9	9	9	9	9	9	10	11
B4. Other flows 3/	9	8	9	9	10	9	9	9	9	10	11
B5. Depreciation	9	9	10	10	10	10	10	10	11	11	12
B6. Combination of B1-B5	9	8	9	10	10	10	9	9	10	10	12
C. Tailored Tests											
C1. Combined contingent liabilities	9	8	9	9	9	9	9	8	9	9	10
C2. Natural disaster	9	8	9	9	9	9	9	9	9	10	10
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	18	18	18	18	18	18	18	18	18	18	18

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Vanuatu: Sensitivity Analysis for Key Indicators of Public Debt 2019–29

	Projections 1/										
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
PV of Debt-to-GDP Ratio											
Baseline	35	36	37	37	38	39	40	41	42	43	44
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	35	35	35	34	34	33	33	33	32	32	32
B. Bound Tests											
B1. Real GDP growth	35	37	40	42	43	45	46	48	50	52	53
B2. Primary balance	35	38	41	41	42	42	43	44	45	46	47
B3. Exports	35	38	43	43	44	45	45	46	47	48	49
B4. Other flows 3/	35	41	46	47	47	48	49	49	50	51	52
B5. Depreciation	35	43	41	40	38	37	37	36	35	35	34
B6. Combination of B1-B5	35	37	38	36	37	37	38	39	40	41	42
C. Tailored Tests											
C1. Combined contingent liabilities	35	42	42	43	43	44	45	46	47	48	49
C2. Natural disaster	35	44	45	47	48	49	50	52	53	55	**56**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Public debt benchmark	55	55	55	55	55	55	55	55	55	55	55
PV of Debt-to-Revenue Ratio											
Baseline	127	129	131	134	137	140	145	149	153	158	162
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	127	125	123	122	121	120	120	119	119	119	119
B. Bound Tests											
B1. Real GDP growth	127	133	142	148	154	160	167	174	182	189	197
B2. Primary balance	127	136	145	148	151	154	158	163	167	171	176
B3. Exports	127	136	153	156	159	162	166	170	174	178	182
B4. Other flows 3/	127	146	165	168	170	173	177	181	186	190	193
B5. Depreciation	127	155	150	145	141	138	136	134	132	131	130
B6. Combination of B1-B5	127	131	135	131	133	137	141	145	149	154	158
C. Tailored Tests											
C1. Combined contingent liabilities	127	149	151	154	157	160	164	168	173	177	181
C2. Natural disaster	127	157	161	166	171	176	182	189	195	201	208
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	11	12	11	12	12	9	8	11	9	10	12
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	11	12	11	11	11	8	8	10	8	9	10
B. Bound Tests											
B1. Real GDP growth	11	12	11	12	12	9	9	12	10	11	13
B2. Primary balance	11	12	11	12	12	9	9	11	10	10	12
B3. Exports	11	12	11	12	12	9	9	11	9	10	12
B4. Other flows 3/	11	12	11	12	12	9	9	12	10	10	13
B5. Depreciation	11	13	13	13	13	10	10	12	11	11	13
B6. Combination of B1-B5	11	12	11	11	11	8	8	11	9	9	12
C. Tailored Tests											
C1. Combined contingent liabilities	11	12	12	12	12	9	9	11	10	10	12
C2. Natural disaster	11	13	12	13	13	10	10	12	10	11	13
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.

1/ A bold value indicates a breach of the benchmark.

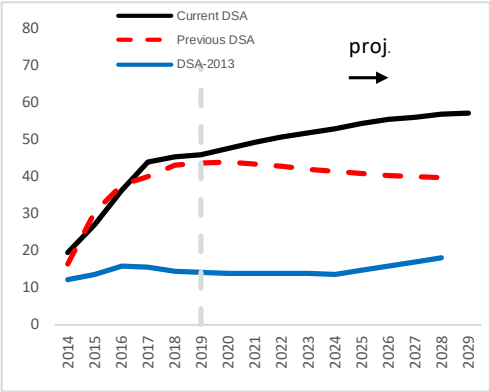
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.

3/ Includes official and private transfers and FDI.

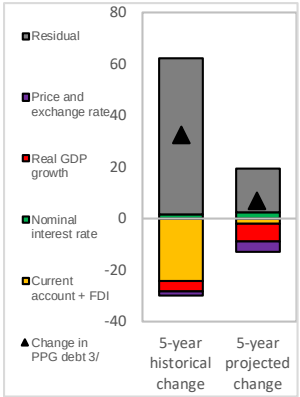
Figure 3. Vanuatu: Drivers of Debt Dynamics – Baseline Scenario

External debt

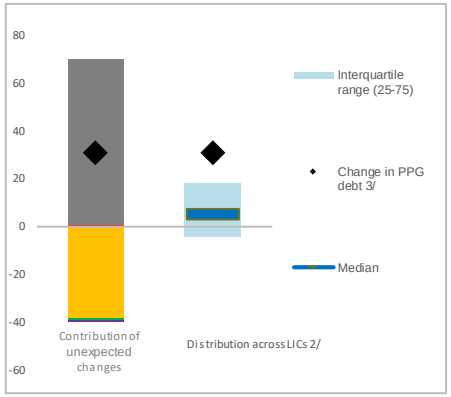
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

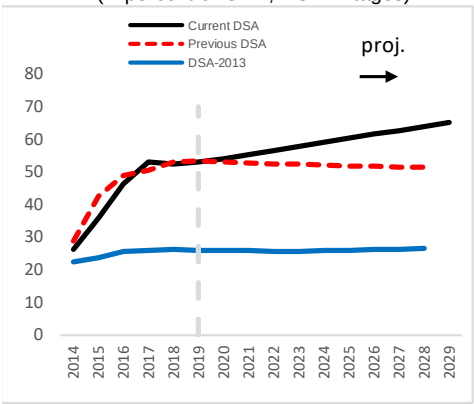


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

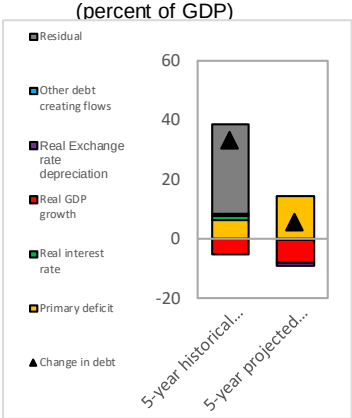


Public debt

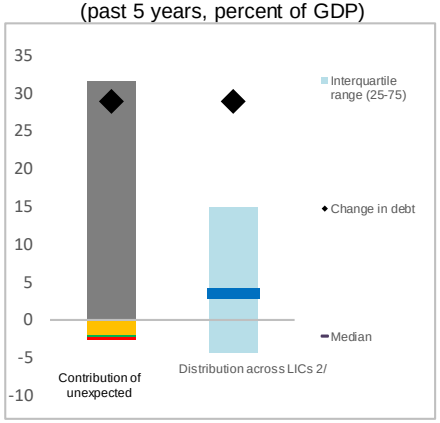
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



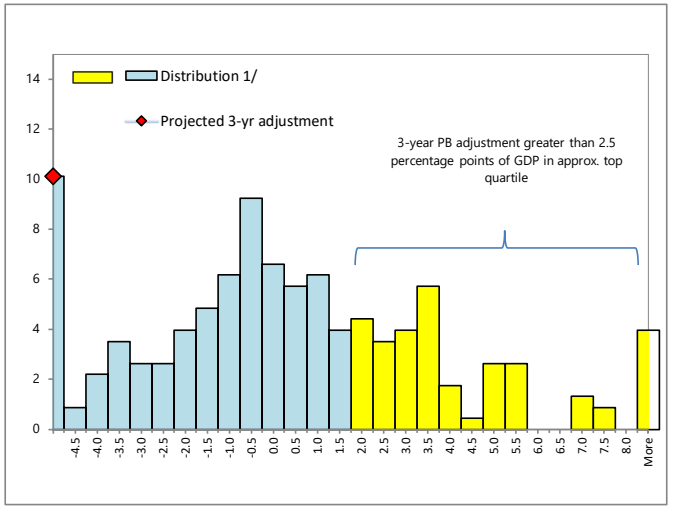
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

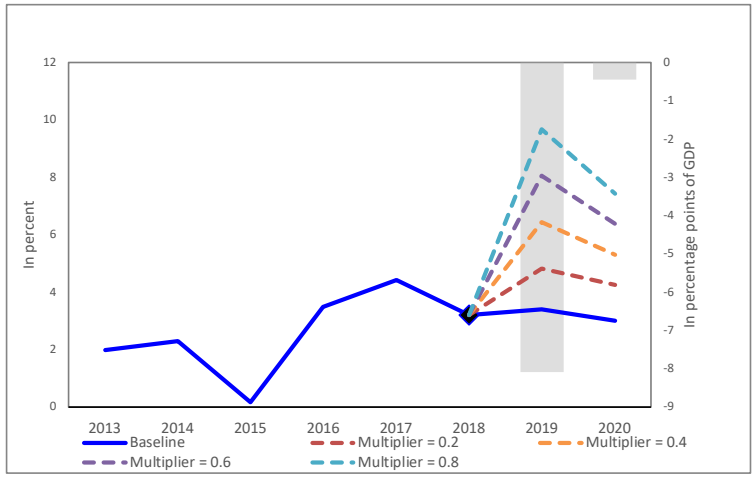
Figure 4. Vanuatu: Realism Tools

3-Year Adjustment in Primary Balance (Percentage points of GDP)



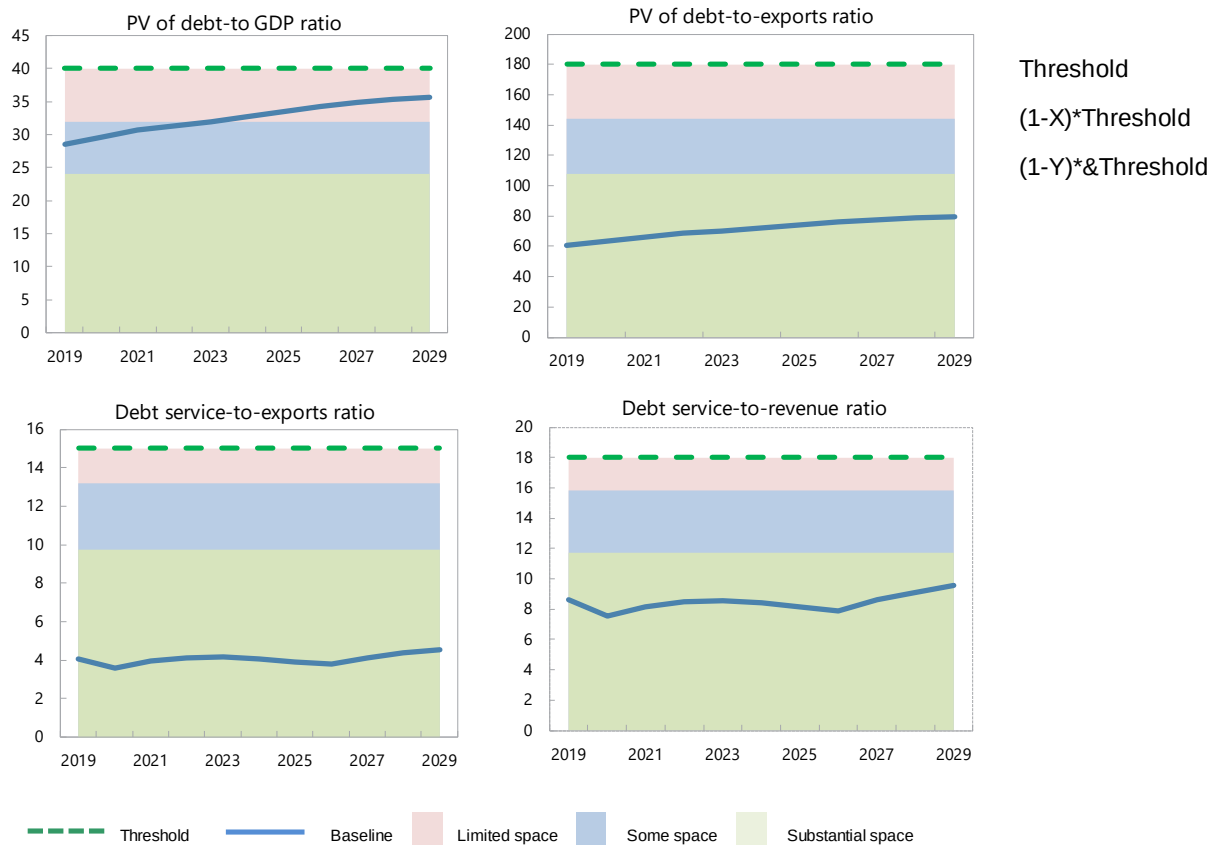
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Figure 5. Vanuatu: Qualification of the Moderate Category, 2019–29 1/



Sources: Country authorities; and staff estimates and projections.

1/ For the PV debt/GDP and PV debt/exports thresholds, x is 20 percent and y is 40 percent. For debt service/Exports and debt service/revenue thresholds, x is 12 percent and y is 35 percent.